

Time permitting (either this week or next): pwnkit

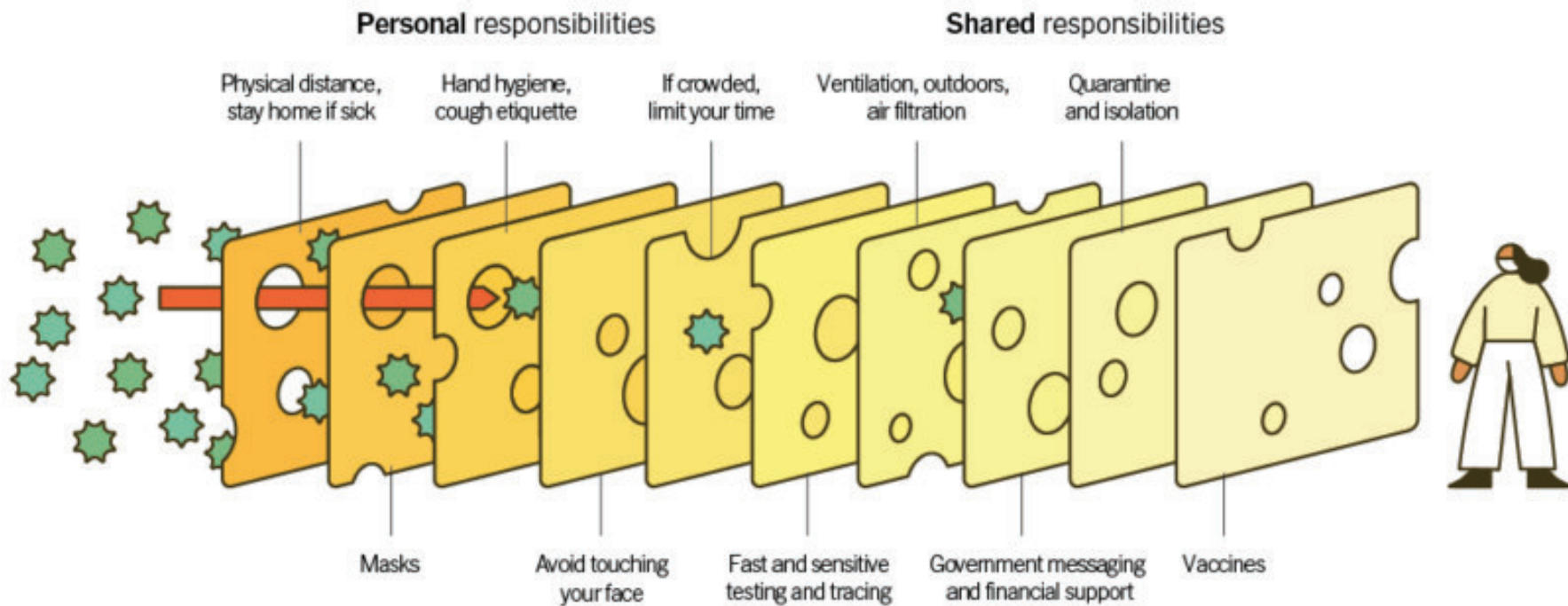
Local privilege escalation vulnerability discovered (2022) in `pkexec` command:

- Bharat Jogi, PwnKit: Local Privilege Escalation Vulnerability Discovered in polkit's `pkexec` (CVE-2021-4034), <https://blog.qualys.com/vulnerabilities-threat-research/2022/01/25/pwnkit-local-privilege-escalation-vulnerability-discovered-in-polkits-pkexec-cve-2021-4034> (last accessed March 6, 2025).
- Can SELinux help? Why or why not? Let's talk about it.

The Return of the Swiss Cheese Analogy! (security tools and techniques are like slices of swiss cheese)

Multiple Layers Improve Success

The Swiss Cheese Respiratory Pandemic Defense recognizes that no single intervention at preventing the spread of the coronavirus is perfect. Each intervention (layer) has holes.



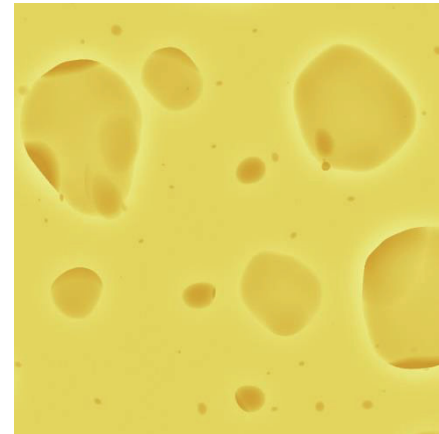
Source: Adapted from Ian M. Mackay (virologydownunder.com) and James T. Reason. Illustration by Rose Wong

Analogy and Take-away

No one security technique is perfect (each has its own holes).

But many used together improves a system's overall security.

- With enough slices, the holes hopefully don't line up!



Understanding the use and limits of each security tool/technique is important

- So: why does pwnkit work on an SELinux system? Can SELinux be used to prevent it?